

Warm-up

- (1) Let $c_n = \frac{3n+2}{4n-1}$. Write out the first four terms, and compute $\lim_{n \rightarrow \infty} c_n$.
- (2) Let $c_n = \frac{\ln(n)}{n}$. Write out the first four terms, and compute $\lim_{n \rightarrow \infty} c_n$.

Medium

- (1) Consider the sequence $(a_n) = (\frac{1}{4}, \frac{2}{5}, \frac{3}{6}, \dots)$. Find a formula for the n th term a_n . Compute $\lim_{n \rightarrow \infty} a_n$.
- (2) Consider the sequence $(a_n) = (6, 2, \frac{2}{3}, \frac{2}{9}, \dots)$. Find a formula for the n th term a_n . Compute $\lim_{n \rightarrow \infty} a_n$.
- (3) Let $b_n = \frac{n!}{(n+2)!}$. Write out the first four terms of the sequence, and find $\lim_{n \rightarrow \infty} b_n$, with justification.
- (4) Let $b_n = \frac{n!}{2^n}$. Compute the first four terms of the sequence, and compute $\lim_{n \rightarrow \infty} b_n$, with justification.
- (5) Consider the sequence $(a_n) = (\frac{1}{2}, \frac{2}{5}, \frac{3}{10}, \frac{4}{17}, \dots)$. Find a formula for the n th term a_n . Compute $\lim_{n \rightarrow \infty} a_n$.

Stretch

- (1) Suppose a and b are positive real numbers. Compute $\lim_{n \rightarrow \infty} \sqrt[n]{a^n + b^n}$.
- (2) Define a sequence recursively by $a_1 = 1$ and $a_{n+1} = 1 + \frac{1}{1+a_n}$. Verify that the monotone convergence theorem applies individually to the sequence of odd terms and to the sequence of even terms, and use this to compute the limit of the sequence.